

Tasks'

1. Open your Windows or Linux system, use the command line to assign a static IPv6 address to your primary network adapter, and verify the configuration using ipconfig (Windows) or ifconfig/ip (Linux).



```
Admin State    State      Type      Interface Name
-----
Enabled        Connected  Dedicated Ethernet 2
Enabled        Disconnected Dedicated McAfee VPN
Enabled        Connected  Dedicated Wi-Fi

C:\Windows\System32>netsh interface ipv6 set address "Wi-Fi" static 2001:db8:1::10

Invalid address parameter (static). It should be a valid IPv6 address.

C:\Windows\System32>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet 2:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::1d7:6a6:c5c4:c31b%8
    IPv4 Address. . . . . : 192.168.56.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.1.1

Unknown adapter McAfee VPN:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Local Area Connection* 2:

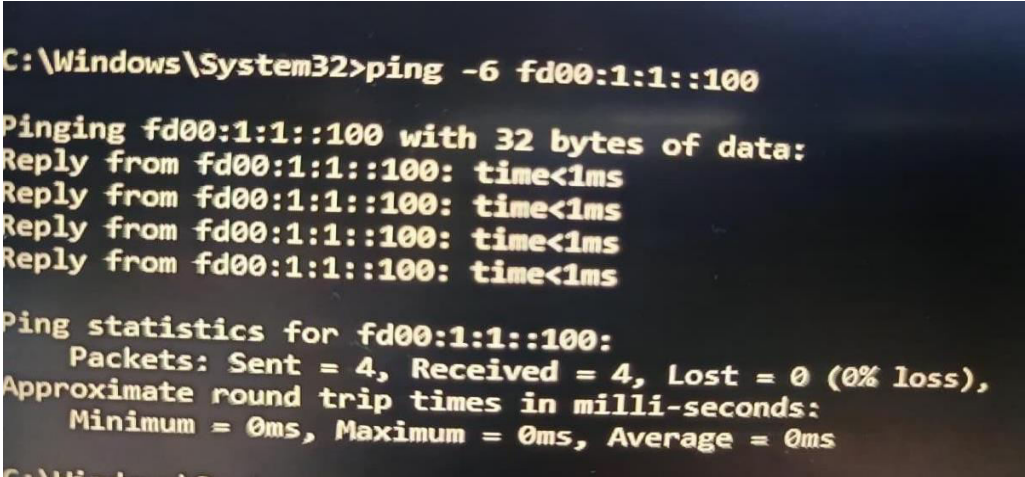
    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . : domain.name
    IPv6 Address. . . . . : 2401:4900:8f73:f34:785f:6e82:3384:c885
    Temporary IPv6 Address. . . . . : 2401:4900:8f73:f34:2082:adc3:254f:46a9
```

2. Configure a unique local IPv6 address (starting with 'fd') on your system and confirm connectivity by pinging another device on your LAN that also has a unique local IPv6 address.

Hint: Use the ping6 or ping -6 command for IPv6 testing.



```
C:\Windows\System32>ping -6 fd00:1:1::100

Pinging fd00:1:1::100 with 32 bytes of data:
Reply from fd00:1:1::100: time<1ms
Reply from fd00:1:1::100: time<1ms
Reply from fd00:1:1::100: time<1ms
Reply from fd00:1:1::100: time<1ms

Ping statistics for fd00:1:1::100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

3. Set up a temporary IPv6 address on your system and check if it changes after a reboot or network disconnect/reconnect.

Constraint: Document the steps and the observed address behavior in a text file.



OLD ADDRESS:-

Temporary IPv6 Address. : 2409:40c1:603e:bf4d:21a3:aa21:9745:9062

NEW ADDRESS:-

Temporary IPv6 Address. : 2409:40c1:603e:bf4d:bde9:a1a2:87ab:c368

4. Simulate a Zomato-style search: Write a small script (any language) that stores an array of restaurant names and their IPv6 addresses, then allows the user to input a partial name and returns the matching restaurant's IPv6 address.



```

# Restaurant names and IPv6 addresses restaurants = {
    "Dominos Pizza": "fd00:1:1::101",
    "Pizza Hut": "fd00:1:1::102",
    "McDonald's": "fd00:1:1::103",
    "Burger King": "fd00:1:1::104",
    "Subway": "fd00:1:1::105"
}

# User input search = input("Enter restaurant name:
").lower()

# Search for matching restaurant found =
False

for name, ipv6 in restaurants.items():    if
search in name.lower():
print("Restaurant:", name)
print("IPv6 Address:", ipv6)        found =
True if not found:
    print("Restaurant not found.")

```

5. Use ChatGPT or Copilot to generate a sample IPv6 subnetting scenario (with at least 3 subnets), and then manually assign a valid IPv6 address from one of those subnets to your system. Paste the AI-generated scenario and your configuration steps in a text file.

IPv6 Subnetting Scenario and Configuration

AI-Generated IPv6 Subnetting Scenario Organization

prefix: fd00:1234:abcd::/48

Three subnets:

1. Admin Network

- Subnet: fd00:1234:abcd:1::/64

2. Staff Network

- Subnet: fd00:1234:abcd:2::/64

3. Guest Wi-Fi

- Subnet: fd00:1234:abcd:3::/64

Assigned Address

I selected the Staff Network.

IPv6 Address: fd00:1234:abcd:2::100/64

Configuration Steps (Windows)

1. Open Command Prompt as Administrator.
2. Check the adapter name: netsh interface ipv6 show interfaces 3.
Assign the IPv6 address: netsh interface ipv6 set address "Wi-Fi" fd00:1234:abcd:2::100
4. Verify: ipconfig